

MATERIALS SCIENCE

A unified ligand-dimensional design to halt cation migration in perovskite photovoltaics

Zhipeng Miao^{1†}, Shiheng Wang^{1†}, Qi Pan^{1†}, Jie Yu², Yapeng Shi¹, Yunhang Xie¹, Yipei Wang², Sihui Peng¹, Ting Zhang¹, Hong Lu^{3,4}, Hainam Do⁴, Yiqiang Zhang^{1*}, Yanlin Song^{5*}, Pengwei Li^{1,2*}

Organic cation migration, particularly that of formamidinium (FA⁺), remains a major factor limiting the long-term stability of perovskite solar cells (PSCs). Low-dimensional/three-dimensional (LD/3D) heterojunctions have shown promise for mitigating this migration, yet systematic comparisons under a unified processing and scaffold framework are lacking. Here, we develop a family of pyridine-based ligands that enable the formation of structurally well-defined LD/3D heterojunctions spanning perovskite-related 2D, 1D, 1D intercalated, and 0D metal-halide motif architectures under identical processing. This platform reveals that the 2D/3D configuration uniquely couples geometric confinement with multidirectional hydrogen bonding to immobilize FA⁺ and enhance carrier performance. Consequently, PSCs achieve a champion efficiency of 26.88% (certified 26.52%) and outstanding stability, retaining >91% initial efficiency after 1000 hours at 85°C/85% relative humidity and >94% after 2000 hours of maximum power point tracking. Large-area modules (900 square centimeters) deliver 20.25% efficiency (certified 17.75%) with negligible degradation over 4 months. This work establishes a ligand-based dimensional design platform for stable, scalable perovskite photovoltaics.

INTRODUCTION

Organic cation migration, particularly that of formamidinium (FA⁺) and methylammonium (MA⁺), critically governs the stability and performance of low-dimensional/three-dimensional (LD/3D) perovskite solar cells (PSCs) (1–3). These highly mobile species readily traverse heterogeneous interfaces, initiating structural degradation and undermining long-term device operation (4, 5). Among them, the irreversible migration of FA⁺ remains especially problematic (6), persisting even as device efficiencies have improved through reversible MA⁺ modulation or cesium (Cs⁺) incorporation strategies that elevate the FA⁺ migration energy barrier (7, 8). LD/3D perovskite heterojunctions, typically engineered through bulk-phase modulation or interfacial reconstruction, incorporate a diverse set of LD phases, including corner-sharing 2D perovskites (9–14), edge-sharing 1D perovskites (15–18), face-sharing 1D intercalated (1D-Inter.) structures (19–21), and isolated octahedral 0D perovskites (22–25), and have enabled notable advances in stability and scalability (3, 14). However, a systematic understanding of ion migration across these dimensionally engineered heterostructures remains elusive for two reasons: (i) the intrinsic complexity of crystallization hampers the reproducible formation of phase-pure, architecture-defined heterojunctions (26–30); and (ii) large variations in spacer geometry and physicochemical properties across reported systems hinder direct, head-to-head comparisons of cation migration dynamics within a unified scaffold and processing window (31–37).

To address these challenges, we designed a family of structurally analogous pyridine-derived ligands (PDs), including APD (4,4'-azobispyridine), DPD (4,4'-dithiodipyrindine), DPE (4,4'-dipyridylethane), and APY (3-aminomethylpyridine). This ligand platform enables controlled crystallization of phase-pure LD/3D heterojunctions with tunable dimensionality, providing an ideal model to investigate how ligand chemistry and geometric confinement jointly regulate cation migration (movie S1). Among these PDs, APD forms a 2D/3D heterostructure that integrates a multidirectional hydrogen bonding network, derived from the molecular design of the ligand, with a layered 2D confinement geometry. This dual mechanism establishes directional chemical interactions with FA⁺ and imposes spatial diffusion barriers, substantially increasing the FA⁺ migration energy barrier relative to the perovskite-related 1D (edge-/face-sharing) and 0D (isolated-octahedra) metal-halide motifs. The resulting lattice stabilization and defect passivation markedly enhance charge transport and suppress ionic drift.

These structural advantages directly translate into superior photovoltaic performance. The optimized PSCs achieve a champion power conversion efficiency (PCE) of 26.88% (certified 26.52%) and maintain more than 95% of their initial efficiency after 2000 hours under ambient conditions and more than 91% after 1000 hours at 85°C/85% relative humidity (RH). Continuous maximum power point tracking (MPPT) confirms more than 94% retention after 2000 hours of operation. The strategy is scalable: A 900-cm² perovskite module delivers a champion PCE of 20.25% (certified 17.75%) with negligible degradation after 4 months of encapsulated storage, underscoring the practical viability of this ligand-dimensional design for industrial-scale optoelectronics.

RESULTS

Ligand design and synthesis of LD motif crystals

To rationally explore the mechanism of cation migration in LD/3D heterojunctions spanning perovskite-related 2D, 1D, 1D-Inter., and 0D metal-halide motif architectures, we established a molecular design

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¹College of Chemistry, Zhengzhou University, Zhengzhou 450001, P. R. China.

²Henan Institute of Advanced Technology, Zhengzhou University, Zhengzhou 450001, P. R. China. ³Yangzhou Precision Systems Incorporated, Yangzhou 225000, P. R. China.

⁴Department of Chemical and Environmental Engineering, University of Nottingham Ningbo China, Ningbo 315100, P. R. China. ⁵Key Laboratory of Green Printing, CAS Research/Education Center for Excellence in Molecular Sciences Institute of Chemistry, Chinese Academy of Sciences (ICCAS), Beijing Engineering Research Center of Nanomaterials for Green Printing Technology National Laboratory for Molecular Sciences (BNLMS), Beijing 100190, P. R. China.

*Corresponding author. Email: yqzhang@zzu.edu.cn (Y.Z.); ylsong@iccas.ac.cn (Y.S.); pwlhappy@zzu.edu.cn (P.L.)

†These authors contributed equally to this work.

framework centered on the conformational dynamics and interaction modes of organic cations (Fig. 1A). Specifically, the dimensional outcomes and ionic blocking capability within this ligand series are primarily governed by two coupled design handles: (i) ligand backbone geometry (rigidity, symmetry, and conformational freedom), which biases the attainable inorganic connectivity under the same processing window; and (ii) the multiplicity and directionality of ligand-framework interactions (e.g., hydrogen bonding and steric confinement), which together influence motif formation, phase purity, and ionic diffusivity. We emphasize that these factors act in concert with processing-dependent effects (e.g., guest inclusion or in situ transformation), and our conclusions are scoped to this coherent pyridine-derived platform and the identical film-processing protocol used here. We designed a set of PDs (Fig. 1B) with systematic variations in bridging moieties: APD is planar, rigid, and symmetric with an N=N bridge; DPD incorporates a twisted S—S bond that

undergoes reductive cleavage under the growth conditions to generate 4-mercaptopyridine (4-MPy) as the active ligand; DPE features a rotatable $-\text{CH}_2-\text{CH}_2-$ linker that increases conformational flexibility; and APY is nonplanar, asymmetric, and strongly dipolar with a single anchoring site. Under the identical processing window, these ligands, together with their in situ transformations and solvent interactions, allow a controlled comparison of how molecular structure and processing jointly influence the resulting octahedral dimensionality. The resulting dimensionality is therefore not governed by ligand rigidity alone but by the combined effects of molecular structure, chemical stability under the processing conditions, possible in situ transformation, and ligand-solvent interactions.

In this work, we use “LD motifs” to collectively denote the LD phases enabled by the PDs, including a corner-sharing 2D Dion-Jacobson (DJ) perovskite, perovskite-related 1D (edge-/face-sharing), and 0D (isolated-octahedra) metal-halide motifs (table S1). For

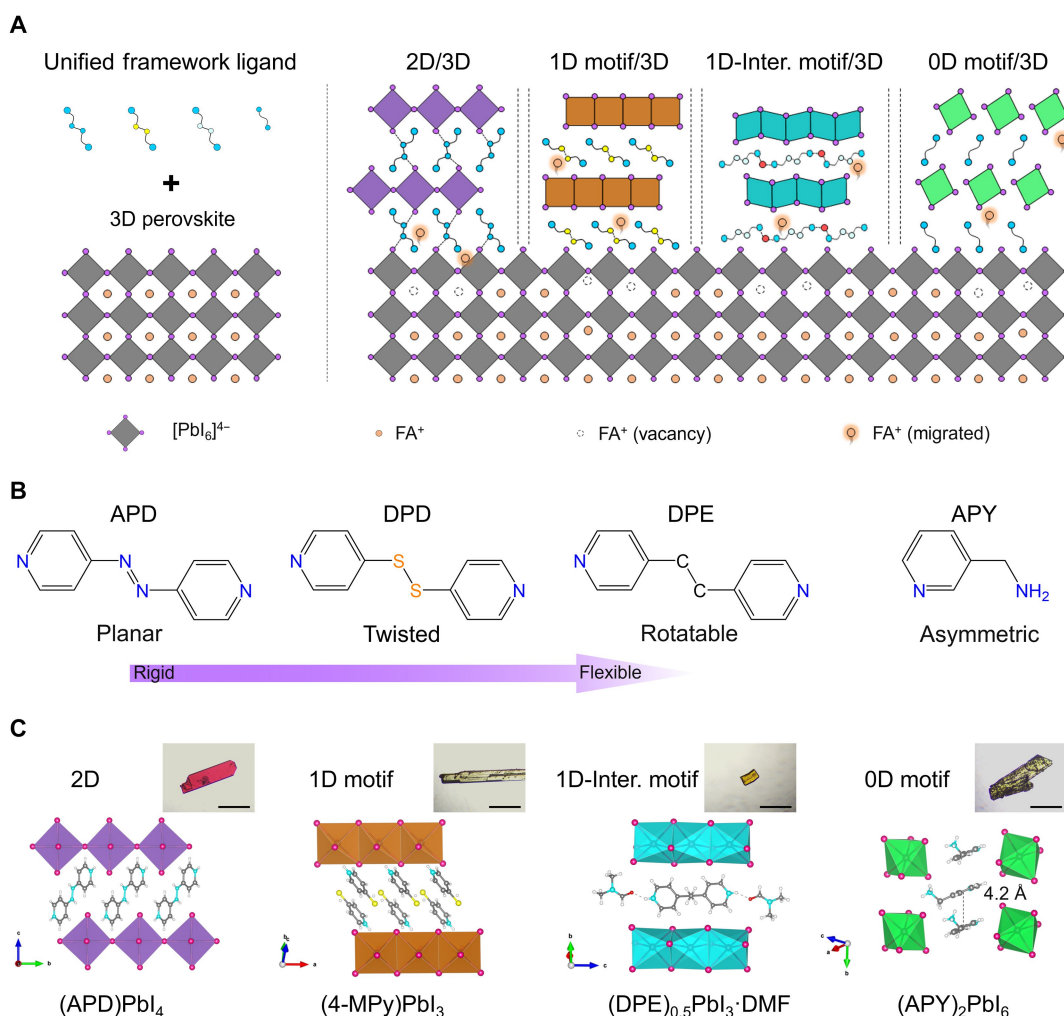


Fig. 1. Schematic diagram of different heterojunctions and crystal structure, optical characterization of LD motif crystals based on PDs. (A) Theoretical model schematics of introducing pyridine derivatives into 3D perovskite (FAPbI₃) via bulk-phase additive engineering for constructing 2D/3D, 1D motif/3D, 1D-Inter. motif/3D, and 0D motif/3D heterojunctions, along with the demonstration of suppressed cation migration. (B) Molecular formula of PDs (APD, DPD, DPE, and APY). The rigidity of the molecule from APD to DPE gradually decreases, and APY is an asymmetric structure. (C) The SCXRD structure and optical microscope images (inset) of (APD)PbI₄, (4-MPy)PbI₃ (derived from in situ reduction of DPD under acidic conditions), (DPE)_{0.5}PbI₃·DMF, and (APY)₂PbI₆ crystals. The APY²⁺ ligands exhibit π - π offset stacking, with a centroid-centroid distance of 4.2 Å. Scale bar, 500 μm . Color code for the atomic species in the crystal structure: pink, iodine (I); gray, carbon (C); light blue, nitrogen (N); red, oxygen (O); yellow, sulfur (S); white, hydrogen (H).

readability, we thereafter refer to them as 2D, 1D motif (edge-sharing), 1D-Inter. motif (face-sharing), and 0D motif. Single crystals of the four LD motifs were grown using an optimized solution-cooling protocol, yielding distinct crystal colors (red, yellow, orange, and light yellow; Fig. 1C, inset). Scanning electron microscopy (SEM) confirmed uniform grain morphology (fig. S1). Single-crystal x-ray diffraction (SCXRD) resolved the expected dimensional frameworks: APD produced corner-sharing 2D DJ layers, DPD underwent reductive cleavage to 4-MPy, yielding a 1D motif (4-MPy)PbI₃, DPE assembled into intercalated 1D face-sharing structures, and APY yielded isolated 0D octahedra stabilized by π - π interactions (Fig. 1C). Powder XRD patterns matched simulated profiles (fig. S2), confirming phase purity. The SCXRD structural details are provided in figs. S3 to S7. Complementary optical characterization, electronic structure analysis, and synthesis parameters are presented in figs. S8 to S10, note S1, and tables S2 and S3.

To elucidate how PDs influence dimensionality, we further examined their protonation-induced structural evolution (figs. S11 to S16 and tables S4 and S5). The protonated APD²⁺ adopts a distorted geometry with four hydrogen bonding sites (dihedral angle: 77.0°; fig. S4C). Meanwhile, the electrostatic potential map reveals a broadly distributed negative surface, enabling multidentate coordination. This feature facilitates the stabilization of the organic-inorganic framework and promotes the formation of well-ordered 2D perovskite. In contrast, the flexible S—S bonds in DPD generate flexible sulfur-containing segments under strongly acidic and reducing growth conditions. The resulting Pb—S coordination provides only limited anchoring geometries, thus restricting multidirectional connectivity and ultimately forming a 1D motif chain-like framework. The flexible —CH₂—CH₂— bridge in DPE promotes strong interactions with *N,N*-dimethylformamide (DMF) during crystallization, generating ligand-solvent complexes that intercalate between [PbI₆]⁴⁻ chains. Acting primarily as spacers rather than multidentate anchors, these complexes disrupt extended 2D order and direct the assembly of high-stable 1D-Inter. motif structures (fig. S17 and note S2). The APY cation is nonplanar and asymmetric, with a strong dipole moment and a single anchoring site, which sterically hinders the formation of extended 1D or 2D networks. Its directional hydrogen bonding and interactions with the solvent further isolate each [PbX₆]⁴⁻ octahedron, resulting in discrete 0D metal-halide clusters. In contrast to conventionally synthesized 2D (APY)PbI₄ (note S3), the discrete 0D motif structure offers a more competitive yield, owing to its superior thermodynamic and environmental stability at room temperature.

Characterization of LD/3D heterojunctions

To systematically elucidate how LD motifs suppress organic cation migration within 3D perovskite films, we fabricated LD/3D heterojunction films by incorporating the four PD ligands. The morphological characteristics of the perovskite films were scrutinized using SEM. As illustrated in Fig. 2 (A to E), characteristic sheet-like features were observed preferentially at grain boundaries, with fewer appearing on the film surface and a distribution previously shown to effectively passivate intergranular and surface defects (22). Notably, the grain size of the LD/3D films exhibited markedly larger than that of the control, suggesting that the addition of PDs effectively modulates the crystallization process and promote superior film formation. This conclusion was further supported by in situ ultraviolet-visible (UV-vis) absorption and photoluminescence (PL) measurements, which revealed that PD ligands dynamically regulate crystal nucleation

and growth processes (fig. S18 and note S4). Cross-sectional SEM imaging confirmed consistent internal morphology (fig. S19), while 3D profilometry analysis revealed reduced surface roughness compared to the control film (fig. S20). Collectively, these morphological improvements translated into enhanced optoelectronic properties: The LD/3D films exhibited stronger light absorption, brighter PL emission (fig. S21), and significantly extended fluorescence lifetimes, which most pronounced in the 2D/3D configuration (fig. S22 and table S6) and consistent with suppressed nonradiative recombination and improved electronic quality. Ultrafast transient absorption (TA) spectroscopy was used to further probe the photophysical dynamics of the control and ligand-programmed films (fig. S23). Compared with the control (fig. S23, A to C), the 2D/3D film exhibited distinct spectral evolution and kinetic behavior (fig. S23, D to F), consistent with interfacial carrier/energy transfer processes in the heterostructure. In particular, the slower bleach recovery dynamics indicated reduced trap-assisted recombination and improved electronic quality, in agreement with the time-resolved PL results.

To probe the internal structural heterogeneity and verify LD motif incorporation, transmission electron microscopy (TEM) was performed (Fig. 2, F to J). The measured lattice spacings for the embedded LD motifs were 11.63 Å (2D), 3.95 Å (1D motif), 11.94 Å (1D-Inter. motif), and 8.89 Å (0D motif), respectively, and were in excellent agreement with single-crystal parameters of the corresponding PD-based perovskites, confirming faithful phase integration. The statistical 2D phase distribution map demonstrated that the 2D/3D system dominated by *n* = 1 2D phase distribution alongside the 3D phase (fig. S24). Complementary XRD analysis of the films (fig. S25) revealed distinct diffraction peaks arising from the embedded LD domains. For example, the 2D/3D film exhibited a prominent new peak at 7.6° 2 θ , matching the (002) reflection of the simulated 2D crystal structure. Similarly, the 1D motif/3D film showed additional peaks at 7.6° and 8.6° 2 θ , corresponding to the (001) and (011) planes of DPD-derived 1D motif. In the 1D-Inter. motif/3D film, new reflections at 8.9° and 9.9° 2 θ assigned to the (110) and (020) planes of the 1D-Inter. motif (DPE)_{0.5}PbI₃·DMF structure. Last, the 0D motif/3D film displayed characteristic peaks at 9.9° and 10.6° 2 θ , consistent with the (002) and (200) reflections of the corresponding 0D motif. Enhanced homogeneity of the 2D/3D film was confirmed by Kelvin probe force microscopy mapping (Fig. 2, K to O). This was evidenced by a more negative contact potential difference (V_{CPD} = -280 mV) and a significant reduction in the contact potential variation (CPV) from 50 mV (3D control) to 12 mV (table S7). The markedly reduced spatial fluctuations and diminished grain boundary potential steps point to a flatter interfacial band landscape, which favors more selective hole extraction and suppresses interfacial recombination. Together, these structural and spectroscopic results demonstrate the successful synthesis of a complete series of phase-pure LD/3D heterojunctions using a unified PD ligand design strategy. The well-resolved structure-morphology-property correlations established here provide a robust foundation for unraveling the mechanisms by which dimensional engineering suppresses organic cation migration.

Cation migration in LD/3D heterojunctions and films

We first probed interfacial chemistry by x-ray photoelectron spectroscopy (XPS). In the 2D/3D film, the Pb 4f binding energy shifts by more than 0.5 eV toward lower energy relative to the pristine 3D control, indicating increased local electron density and stronger interfacial bonding (Fig. 3A). Films based on 1D, 1D-Inter., and 0D

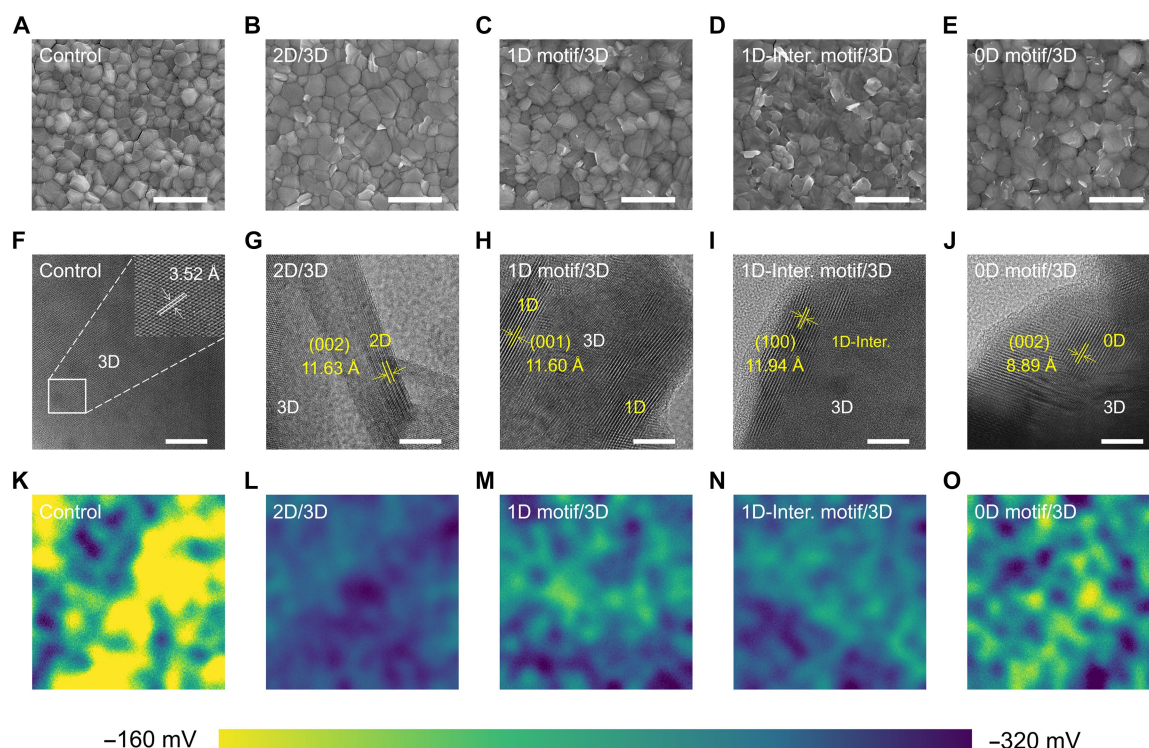


Fig. 2. Characterization of 3D and LD/3D heterojunction films. (A to E) SEM images of 3D, 2D/3D, 1D motif/3D, 1D-Inter. motif/3D, and 0D motif/3D films. Scale bars, 1 μm . SEM characterization was performed under high vacuum with an acceleration voltage of 10 kV. Thin gold layers were sputter-coated onto the sample surfaces before imaging to enhance their conductivity. (F to J) TEM images of 3D, 2D/3D, 1D motif/3D, 1D-Inter. motif/3D, and 0D motif/3D films. Scale bars, 10 nm. For TEM analysis, the film samples were gently scraped into fine powder and then dispersed onto the copper grid. The lattice spacing of the LD motifs is highlighted by the yellow signals. (K to O) Surface potential variation of 2D/3D, 1D motif/3D, 1D-Inter. motif/3D, and 0D motif/3D films. The color scale represents the surface potential, with values ranging from -160 to -320 mV, as indicated by the color bar.

motifs show smaller shifts (0.31, 0.31, and 0.21 eV, respectively), consistent with weaker steric and electronic interactions between non-2D ligands and the $[\text{PbI}_6]^{4-}$ framework. These surface signatures align with the PL trends, in which the 2D/3D film exhibits the highest intensity, indicating suppressed nonradiative recombination at grain boundaries. Because long-term stability is governed by bulk cation dynamics, we focus on FA^+ migration, which is dominant under operation, and we note that MA^+ and Cs^+ migrate more reversibly and to smaller extents.

To quantify migration energetics, we built atomistic models of 3D FAPbI_3 and four LD/3D heterojunctions (2D, 1D motif, 1D-Inter. motif, and 0D motif), and we computed minimum energy pathways for FA^+ (figs. S26 and S27). In 3D FAPbI_3 , FA^+ readily traverses the Pb—I bottleneck under thermal excitation with a relatively low barrier of 8.29 eV, which is consistent with its high susceptibility to irreversible ion loss (Fig. 3B and fig. S26A). By contrast, the 2D/3D heterostructure exhibits a two-stage suppression mechanism, namely, geometric confinement by the layered LD motif framework and strong hydrogen bonding between protonated PD ligands and FA^+ . This combination raises the barrier to 17.52 eV, which exceeds those of 1D motif/3D (15.13 eV), 1D-Inter. motif/3D (15.75 eV), and 0D motif/3D (15.86 eV) (fig. S27, D to F). These climbing-image nudged elastic band (CI-NEB) values do not represent single-ion migration barriers and are not directly comparable in absolute magnitude to experimentally measured activation energies. Instead, they reflect the total energy change associated with coordinated migration in the

simulated supercell, including lattice relaxation and interfacial structural response, and are therefore used here primarily for relative comparison across different heterojunction architectures. Similar suppression is found for MA^+ in 2D/3D heterojunctions (fig. S28). Temperature-dependent electrochemical impedance spectroscopy (EIS) further corroborates this trend, because the 2D/3D device shows a larger ion migration activation energy $E_a = 0.88$ eV than the others (0.48 eV), which is consistent with the results of theoretical calculations (Fig. 3C). Full details about temperature-dependent EIS are presented in figs. S29 and S30, tables S8 to S9, and note S5. Variable-temperature solid-state nuclear magnetic resonance (ssNMR) measurements support reduced cation mobility. In 3D FAPbI_3 , FA^+ and MA^+ exhibit temperature-enhanced dynamic proton signals indicative of high mobility (Fig. 3D). In the 2D/3D film, broadened yet intensity-stable proton peaks point to restricted rotational freedom and suppressed thermal motion due to hydrogen bonding and steric confinement (Fig. 3E, fig. S31, and note S6).

Last, device-level imaging under accelerated aging (85°C) visualizes the suppression of cation drift. Time-of-flight secondary ion mass spectrometry (TOF-SIMS) shows pronounced FA^+ redistribution toward the hole transport layer (HTL) and bidirectional MA^+ drift in pristine 3D films (Fig. 3F and fig. S33A), whereas 2D/3D devices display negligible FA^+ displacement and a uniform cation profile through the thickness (Fig. 3, G and H). Other LD motif architectures (1D, 1D-Inter., and 0D motifs) yield only partial suppression, with residual interfacial accumulation (fig. S32). MA^+ mobility is

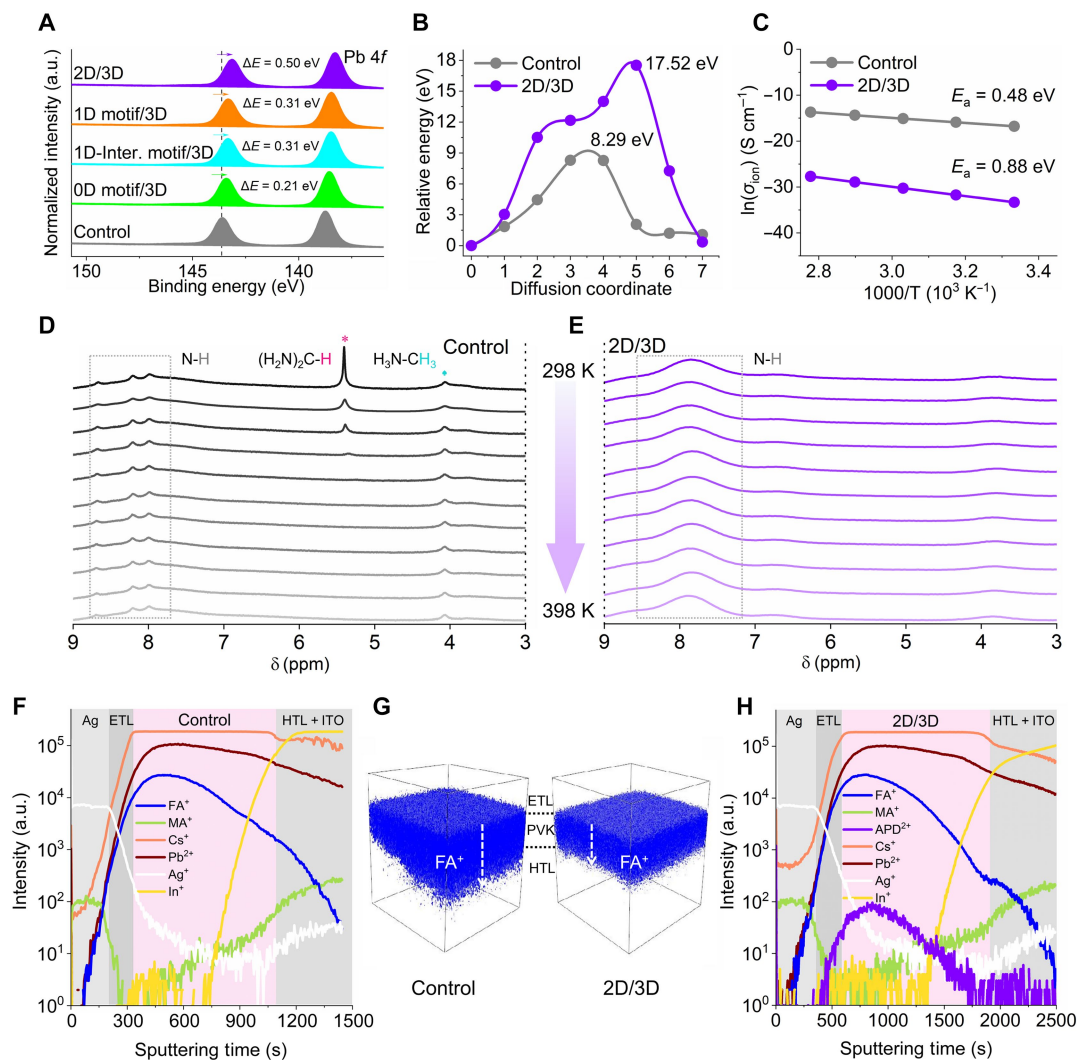


Fig. 3. Cation migration energy barrier based on 3D and 2D/3D heterojunction. (A) XPS of Pb 4f based on 3D, 2D/3D, 1D motif/3D, 1D-Inter. motif/3D, and 0D motif/3D films. (B) Ion mobility activation energy of FA⁺ in 3D and 2D/3D perovskites calculated on the basis of DFT. Initialize an elastic band containing six intermediate state images along the possible migration path of FA⁺, with the initial and final configurations fixed as the initial and final states. (C) Plots of $\ln(\sigma_{ion})$ versus $1000/T$ for five device architectures measured at 300, 315, 330, 345, and 360 K. Lines are ordinary least squares fits with error-based weights and slopes give the activation energy E_a according to $\sigma_{ion} = \sigma_0 \exp(-E_a/k_B T)$. (D and E) The variable-temperature ssNMR spectra were acquired for both the 3D and 2D/3D heterojunction films over a temperature range of 298 to 398 K. The results reveal significant temperature-dependent changes in the cationic chemical environments within the 3D film. The 2D/3D film exhibits broadened signal peaks and low temperature sensitivity due to its robust hydrogen bond network, which enhances structural stability and suppresses cation migration. (F) TOF-SIMS of heat-aged 3D perovskite photovoltaic device. Color coding: white (Ag⁺), light orange (Cs⁺), burgundy (Pb²⁺), blue (FA⁺), mint green (MA⁺), and yellow (In⁺). Unencapsulated devices were aged at 85°C for 24 hours. (G) FA⁺ distribution in 3D and 2D/3D PSCs reveals heterogeneous migration patterns. PVK, perovskite(s). (H) TOF-SIMS of heat-aged 2D/3D perovskite photovoltaic device. a.u., arbitrary units; ppm, parts per million.

attenuated in all LD/3D films but most strongly in 3D control (fig. S33), although MA⁺ and FA⁺ exhibit different migration scales (note S7). Depth-resolved mapping further verifies that PD ligands are homogeneously embedded across all LD/3D heterojunctions without interfacial aggregation or asymmetric enrichment (figs. S34 to S38 and note S8). The spatial distribution of organic cations (fig. S39) indicates that the LD motifs are preferentially localized at grain boundaries and film surfaces rather than forming continuous interfacial layers, consistent with their role as localized ion-blocking domains in the LD/3D heterojunctions. Further TOF-SIMS depth profiling after MPPT at 85°C reveals that both key cations and anions (I⁻) in

the 2D/3D system exhibit significantly suppressed migration signals (figs. S40 and S41). Across the same device architecture and processing, the motif-dependent ion blocking is consistently supported by a converged evidence chain: interfacial migration barriers from density functional theory (DFT), activation energies from temperature-dependent EIS, and direct ionic redistribution from depth profiling under relevant stress conditions.

Performance and stability of PSCs

To assess the practical impact of the LD/3D framework on device performance, an inverted PSC with an indium tin oxide (ITO)/HTL/

perovskite/ETL (electronic transport layer)/Au architecture was fabricated. A schematic of the device is presented in Fig. 4A. Typical J - V curves for 3D and LD/3D devices are shown in Fig. 4B and fig. S42. The open-circuit voltages (V_{OC}) of the control, 2D/3D, 1D motif/3D, 1D-Inter. motif/3D, and 0D motif/3D devices were 1.17, 1.20, 1.17, 1.17, and 1.18 V, respectively (table S10). The integrated photocurrents calculated by the external quantum efficiency (EQE) of the control and modified groups were 25.53, 25.88, 25.58, 25.85, and 25.56 mA cm^{-2} , respectively, which were closely matched the short-circuit currents (J_{SC}) measured by J - V (fig. S43). The corresponding champion PCEs were 26.88% (2D/3D), 25.65% (1D motif/3D), 25.94% (1D-Inter. motif/3D), and 25.70% (0D motif/3D), compared with 25.61% for the control. The champion device maintained a steady-state

output PCE of $\sim 26.7\%$ under a bias of 1.05 V for 300 s, outperforming the control device (fig. S44). The 2D/3D device was independently certified at 26.52% with $V_{OC} = 1.192$ V, $J_{SC} = 26.48$ mA cm^{-2} , and fill factor (FF) = 84.00% under reverse scan (fig. S45). Statistics for 15 fresh devices confirm superior reproducibility (Fig. 4C, fig. S46, and table S11).

To assess scalability, we fabricated 30 cm-by-30 cm modules. The best module reached 20.25% by in-house measurement (Fig. 4, D and E). The steady-state power output at the MPPT for the champion 2D/3D modules was 19.5% (fig. S47), which agreed with the PCE value derived from the J - V curve. Then, a 17.75% PCE was certified under AM1.5G (fig. S48). This certified value ranks among the highest reported for large-area modules of comparable size (fig. S49).

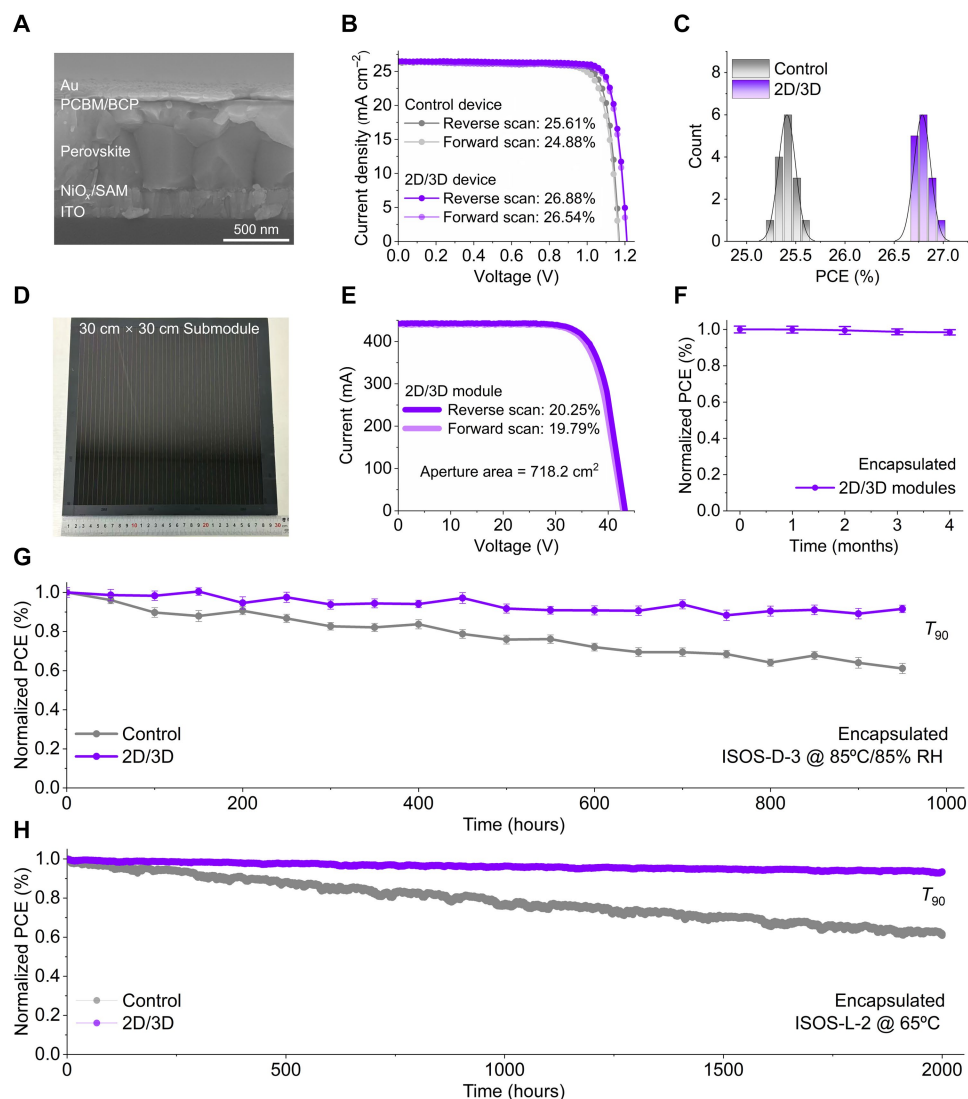


Fig. 4. Performance and long-term operational stability of PSCs and modules. (A) Cross-sectional SEM image of an inverted PSC device (Au/ETL/perovskite/HTL/ITO/glass). Scale bar, 500 nm. (B) J - V plots of the 3D and 2D/3D PSC devices (effective area: 0.0755 cm^2). The PSC devices were tested under the condition of AM1.5G, 1000 W m^{-2} , 25.0°C . (C) PCE distribution of the 15 fresh devices based on the 3D and 2D/3D PSC devices. (D and E) Photograph and J - V curves of the champion 30 cm-by-30 cm perovskite photovoltaic module based on optimization strategies. Perovskite photovoltaic modules were tested under standard test conditions: total irradiance of 1000 W m^{-2} , temperature of 25.0°C , RH of 37%, and spectral distribution of AM1.5G. (F) The PCE evolution of the encapsulated perovskite photovoltaic module in ambient air. The perovskite photovoltaic modules were tested collectively at monthly intervals. (G) Normalized PCE evolution of the encapsulated devices measured under $85^\circ\text{C}/85\%$ RH. (H) Normalized PCE evolution of the encapsulated devices measured at MPPT under the condition of AM1.5G, 1000 W m^{-2} at 65°C in ambient air.

Encapsulated modules showed negligible degradation over 4 months of ambient storage (Fig. 4F).

Charge-transport analyses support the observed performance trends. Space-charge-limited current measurements indicate reduced trap densities in ligand-modified devices, as extracted from trap-filled limit voltage (V_{TFL}) (fig. S50 and table S12). Mott-Schottky analysis showed increased built-in potentials (V_{bi}) in LD/3D devices due to favorable energy level alignment, with 2D/3D devices exhibiting the highest V_{bi} (fig. S51). Transient photocurrent (TPC) shows faster decay, consistent with improved extraction, while transient photovoltage (TPV) decays more slowly, indicating suppressed trap-assisted recombination (fig. S52, A and B, and table S13).

Stability was examined from film to device. LD/3D films display larger and more persistent contact angles than 3D references, indicating enhanced surface hydrophobicity (fig. S53). Unencapsulated devices stored in air retain >95% of initial PCE after 2000 hours, whereas the control drops below 40% (fig. S54). The 2D/3D devices retained 91% of their initial performance after nearly 1000 hours of damp heat testing at 85°C/85% RH, a condition designed to accelerate degradation related to moisture and thermal stress (Fig. 4G). Under MPPT, 2D/3D devices retain 94% after 2000 hours, while controls lose ~40% (Fig. 4H and fig. S55). This comparison with traditional butylammonium (BA)-based devices under identical conditions confirms that APD-based 2D/3D configuration still achieves higher stability (fig. S56). To further clarify the distinction from representative LD strategies and to provide a transparent benchmark, we summarize key differences in ligand scaffold continuity, accessible motif space, processing compatibility, and stability validation in table S14. Beyond these metrics, the superior performance of the APD-based 2D/3D architecture arises from the synergy of its large dipole moment, rigid backbone, well-ordered layer arrangement, and multidirectional hydrogen bonding network, enhancing geometric confinement, electronic coupling, and structural stability.

DISCUSSION

By providing a single-family, multimotif platform under identical processing, this study complements existing LD strategies by enabling an internally controlled comparison of motif-dependent ion dynamics and device-relevant stability, rather than optimizing one spacer concept at a time. Under identical growth conditions, this unified platform enables precise control of dimensionality from 2D perovskite to 0D motifs and systematic tuning of crystallization, film morphology, and optoelectronic properties. Within this framework, the APD-based 2D/3D architecture is particularly effective: Multidirectional hydrogen bonding combined with geometric confinement forms a dual cation-blocking mechanism that suppresses FA^+ migration. As a result, devices reach a certified PCE of 26.52%, maintaining 91% of their initial efficiency after 1000 hours under 85°C/85% RH (ISOS-D-3) and 94% after 2000 hours under the MPPT (ISOS-L-2). Large-area modules (900 cm^2) achieve 17.75% certified efficiency with negligible degradation over 4 months, underscoring scalability and durability.

The mechanistic link between ligand structure, hydrogen bonding networks, and ion migration barriers provides useful guidance for designing LD/3D heterojunctions with improved operational stability. Within the present pyridine-derived platform, our comparison shows that the 2D/3D architecture is the most effective among the examined motifs for suppressing FA^+ migration because it combines

geometric confinement with multidirectional hydrogen bonding. This finding should not be interpreted as a universal rationale for excluding 1D or 0D systems; instead, it suggests that dimensionality selection should be guided by the demands of the target application. More broadly, our results show that ligand design and processing conditions must be considered together when constructing LD/3D heterojunctions and that a unified processing window can serve as a useful basis for controlled comparison and processing-aware materials design. In this sense, the present study offers a conceptual basis for more rational ligand selection under a defined processing window, rather than a fully predictive rule set based solely on ligand characteristics.

MATERIALS AND METHODS

Materials

Hydroiodic acid (HI) (57 wt %) was purchased from Alfa Aesar. APD, DPD, DPE, and APY were purchased from Aladdin. Methylammonium iodide (MAI) (99.5%) was purchased from Xi'an Baolite Optoelectronics Technology. PbI_2 (99.999% perovskite grade), formamidinium iodide (FAI) (99.5%), PbBr_2 (99.5%), NiO_x , methylammonium bromide (MABr) (99.5%), phenethylammonium chloride (PEACl) (99.5%), and ITO (6 $\Omega \text{ sq}^{-1}$) were all purchased from Liaoning Youxuan Technology Co. Ltd. Self-assembled monolayer (SAM) (MeO-4PACz) was purchased from Shanghai Weizhu Chemical Technology Co. Ltd. CsI (99.999%), DMF, 2-methoxyethanol (2-ME), and dimethyl sulfoxide (DMSO) were purchased from Sigma-Aldrich. Ag (99.999%) and Au (99.999%) were purchased from Shanghai Tim New Material Technology Co. Ltd. Ethanol (99.5%), isopropanol (99.5%), and hypophosphorous acid (50 wt % in H_2O) were purchased from Aladdin. Acetone (AR) and anhydrous ether (AR) were purchased from Sinopharm Chemical Reagent Co. Ltd.

Synthesis of (APD) PbI_4

PbI_2 (103.7 mg, 0.225 mmol) was dissolved in a mixture of HI (5 ml) and hypophosphorous acid (400 μl). Then, APD (20.73 mg, 0.1125 mmol) was added and stirred (125°C) until completely dissolved. Subsequently, the red crystal was obtained by slowly lowering to room temperature (5°C hour^{-1}). Last, the crystal was repeatedly rinsed with ether. The crystal was vacuum dried at a constant temperature (65°C) for 10 hours and stored in a glove box (N_2).

Synthesis of (4-MPy) PbI_3

PbI_2 (103.7 mg, 0.225 mmol) was dissolved in a mixture of HI (5 ml) and hypophosphorous acid (400 μl). Then, DPD (50 mg, 0.225 mmol) was added and stirred (120°C) until completely dissolved. Subsequently, the yellow crystal was obtained by slowly lowering to room temperature (5°C hour^{-1}). Last, the crystal was repeatedly rinsed with ether and stored in a glove box (N_2).

Synthesis of (DPE) $_{0.5}\text{PbI}_3$ -DMF

PbI_2 (103.7 mg, 0.225 mmol) and DPE (40 mg, 0.225 mmol) were dissolved in 2 ml of a DMF/DMSO (7/3) mixture. The solution was sealed in an N_2 glove box, heated to 80°C, and stirred for 1 hour. The hot supernatant was then transferred to a new bottle and allowed to cool to room temperature. The resulting crystals were stored in the mother liquor.

Synthesis of (APY) $_2\text{PbI}_6$

PbI_2 (103.7 mg, 0.225 mmol) was dissolved in a mixture of HI (5 ml) and hypophosphorous acid (400 μl). Then, APY (24.4 mg, 0.225 mmol)

was added, heated, and stirred (120°C) until completely dissolved. Subsequently, it was naturally cooled to room temperature to obtain yellow crystal. Last, the crystal was repeatedly rinsed with ether. The crystal was vacuum dried at a constant temperature (65°C) for 10 hours and stored in a glove box (N₂).

Preparation of LD/3D active layer

PbI₂ (1006.85 mg, 2.184 mmol), FAI (342.95 mg, 1.995 mmol), PbBr₂ (38.54 mg, 0.105 mmol), MACl (20.21 mg, 0.299 mmol), MABr (11.76 mg, 0.105 mmol), and CsI (23.39 mg, 0.09 mmol) were dissolved in DMF/DMSO (1152 μ l/348 μ l) mixed solvents. The completely dissolved precursor solution was filtered after standing overnight. The filtered perovskite precursor solution was spin-coated at 4000 rpm for 40 s and rinsed with isopropyl alcohol (IPA) (1 mg ml⁻¹ PEACl) antisolvent for the last 8 s. Subsequently, it was transferred to a 100°C hot stage for annealing for 45 min (in N₂). (APD)I₂ (0.25 mg ml⁻¹ perovskite precursors), (4-MPy)I (0.50 mg ml⁻¹ perovskite precursors), (DPE)I₂ (0.25 mg ml⁻¹ perovskite precursors), and (APY)I₂ (0.20 mg ml⁻¹ perovskite precursors) were added to the experimental group.

Preparation of PSC devices

First of all, the ITO substrate was repeatedly scrubbed by a cotton swab. Then, the scrubbed substrate was placed in a beaker and sequentially sonicated with soapy water, ultrapure water, acetone, and isopropanol for 20 min. The substrate was soaked in ethanol for preservation. When used, the front and back sides of the substrate were dried by N₂ and washed in the UV ozone cleaner for 15 min. Nickel oxide (NiO_x, 15 mg) was dissolved in H₂O (3 ml) and ultrasonically shaken for 5 min until completely dissolved. The filtered NiO_x solution was spin-coated at 1500 rpm for 30 s on a substrate and transferred to a 150°C hot stage annealed for 10 min (in air). Then, the filtered SAM precursor (0.5 mg ml⁻¹ ethanol) was spin-coated at 3000 rpm on a NiO_x layer for 30 s and transferred to a 100°C hot stage for annealing for 60 min (in N₂). Subsequently, the perovskite active layer was applied to the SAM layer. The filtered PC₆₁BM precursor solution (20 mg ml⁻¹ chlorobenzene) was spin-coated at 1500 rpm for 40 s on the perovskite layer and transferred to a 90°C hot stage for annealing for 10 min. Bathocuproine (BCP) precursor solution (0.5 mg ml⁻¹ IPA) was added to the PC₆₁BM layer and spin-coated at 1000 rpm for 60 s (in N₂). Last, the Au (80 nm) electrode was vacuum-evaporated (0.2 to 0.5 Å s⁻¹).

Fabrication of perovskite solar module

Perovskite modules (30 cm by 30 cm) were fabricated on ITO substrates with monolithic series interconnection defined by P1/P2/P3 laser scribing (1064 nm), yielding a geometric fill factor of 97.88%. After P1, substrates were cleaned, and a NiO_x layer was deposited by magnetron sputtering. A SAM was slot-die coated (0.5 mg ml⁻¹, 15 mm s⁻¹, ~60- μ m gap) and annealed at 100°C for 10 min. The perovskite ink (FAI/PbI₂/MACl = 1.0/1.0/0.35 M in 2-ME/DMSO) with (APD)I₂ additive was slot-die coated (20 mm s⁻¹, ~60- μ m gap) at room temperature, followed by vacuum-assisted drying (<10 Pa, 1 min) and annealing at 100°C for 30 min in ambient air. Lithium fluoride (LiF) and C₆₀ were thermally evaporated, SnO₂ was deposited by ALD, and P2 lines were defined. Last, 20-nm ITO and 80-nm Cu were sputtered, and P3 lines were laser-scribed to complete sub-cell interconnection.

Characterization of perovskite films

Optical microscopic photographs of perovskite crystals were taken by a metallographic microscope (Beijing Cossim Scientific Instrument Co. Ltd.). SEM images were collected using the Regulus 8100 (Hitachi High-Tech Co. Ltd.). The XRD patterns were recorded on the Bruker x-ray diffractometer using Cu K α (λ = 1.54056 Å). UV-vis absorption spectra were measured by a Cary 5000 spectrophotometer (Agilent Technologies). The PL and time-resolved PL measurements were conducted by a fluorescence spectrometer (HORIBA FluoroMax). Excitation light at a wavelength of 405 nm was used to test perovskite film samples. All characterizations were performed under room temperature conditions. The 3D profilometry measurements were performed using a Sensofar optical profilometer, a high-resolution noncontact metrology system capable of subnanometer vertical resolution, with a scanning area of 850.08 μ m by 709.32 μ m and a magnification of $\times$ 20.

TA characterization

TA measurements were performed using a Yb:KGW laser system (1030 nm, 60 kHz; Light Conversion) coupled to an optical parametric amplifier to generate the pump pulses. Unless otherwise specified, the samples were excited at 405 nm. A white-light continuum probe (550 to 950 nm) was generated by focusing the 1030-nm beam into an undoped yttrium aluminum garnet (YAG) crystal. The pump-probe delay was controlled by a motorized delay stage. The transmitted probe was collected and analyzed using a fiber-coupled spectrometer equipped with complementary metal-oxide semiconductor sensors (Femto-TA100, Time-Tech Spectra). The pump beam was mechanically modulated at 5 kHz, and the differential absorbance (ΔA) was calculated from consecutive pump-on/pump-off probe spectra. All measurements were conducted at room temperature in ambient air. The pump fluence was kept low (typically on the order of $\sim$ 5 μ J cm⁻²) to avoid nonlinear excitation and photodegradation; the measurements were additionally verified to be in the linear regime by reducing the pump intensity and observing unchanged spectral features and kinetics.

In situ annealing testing of perovskite films

In situ UV-vis and PL were conducted by the laser equipment of Guangzhou Jingyi Optoelectronics Technology Co. Ltd. Before the test, the laser emitting equipment and the hot plate (100°C) under the laser should be built and debugged. The perovskite films were spin-coated in a N₂ glove box and quickly transferred to an in situ test hot plate after spin coating. Parameters were collected at the end of the perovskite films to stabilize the absorption and emission signals.

Temperature-dependent EIS characterization of perovskite films

Temperature-dependent EIS was performed on completed perovskite devices (active area, 0.09 cm²) in the dark under dry N₂, using a 10-mV root mean square (rms) small-signal perturbation around open circuit and sweeping 10⁶ to 10⁻¹ Hz (6 to 10 points per decade) at stabilized temperatures of 300, 315, 330, 345, and 360 K ($\pm$ 0.2 K). Spectra were fitted by complex nonlinear least squares over the full complex plane with a two-arc series model, $Z(\omega) = R_s + [(R_p \parallel CPE_p) + (R_{ion} \parallel CPE_{ion})]$, where constant-phase elements obey $Z_{CPE} = [Q(j\omega)^n]^{-1}$ ($0.5 \leq n \leq 1$); weighting $\omega = 1/|Z|^2$, Kramers-Kronig checks, and parameter stability tests ensured robust identifiability. The low-frequency ionic branch resistance R_{ion} (T) was converted to bulk ionic conductivity via $\sigma_{ion}(T) = t/[R_{ion}(T)A]$

using the measured film thickness t and $A = 0.09 \text{ cm}^2$. Uncertainties from fitting and geometry were propagated to $\ln\sigma$. Activation energies were obtained by Arrhenius analysis of $\ln\sigma_{\text{ion}}$ versus $1/T$ using ordinary least squares with error-based weights, $\sigma_{\text{ion}} = \sigma_0 \exp(-E_a/k_B T)$, taking $E_a = -k_B \times \text{slope}$ and σ_0 from the intercept.

Variable-temperature ssNMR characterization of perovskite films

The variable-temperature ssNMR experiments were conducted on a JNM-ECZL600G spectrometer operating at 599.57 MHz for ^1H , using a 3.2-mm HX double-resonance magic angle spinning (MAS) probe. For sample preparation, the perovskite films were first spin-coated and then carefully scraped into fine powders to ensure optimal packing in the NMR rotor. The measurements used a single-pulse excitation sequence with the following parameters: MAS rate of 15 kHz to minimize spinning sidebands, a relaxation delay of 0.5 s, and 512 scans accumulated to achieve a sufficient signal-to-noise ratio. The temperature-dependent study covered a range from 298 to 398 K, enabling systematic investigation of cation dynamics and phase transitions in these hybrid perovskite materials.

Characterization of PSC devices of TOF-SIMS

The TOF-SIMS were conducted by ION-TOF SIMS 5 (sputtering source: oxygen source, sputtering energy: 1 keV, sputtering area: $250,000 \mu\text{m}^2$, mode: cationic mode). The TOF-SIMS test was performed at room temperature in air, and the devices were placed on an 85°C hot stage for 24 hours before testing. The depth sputtering velocity remains consistent. Furthermore, after encapsulated devices underwent MPPT operation at 85°C for 500 hours, TOF-SIMS was used to analyze the resulting cation redistribution within the devices.

Characterization of PSC devices

The J - V test was conducted using a B2901A source meter equipped with a solar simulator (Enlitech, SS-X50) under AM1.5G. The setting of test parameters was -0.1 to 1.25 V , 0.0755 cm^2 , forward and reverse sweep. The EQE of devices was measured by quantum efficiency measuring instrument (QE-R of Enlitech). TPC, TPV, and EIS were measured in a dark environment by a solar battery carrier characteristic analyzer (PAIOS of Fluxim). Steady-state PCE of the control device and the optimized device were measured using SS-X50 (Enlitech) at 1.02 and 1.05 V for 300 s, respectively.

Characterization of module and device stability

The encapsulated modules were stored long-term under ambient conditions ($25^\circ \pm 3^\circ\text{C}$, atmospheric environment), with periodic performance evaluations. For storage stability (ISOS-D-1), the unpackaged perovskite devices were stored in a temperature-controlled humidity box. Parameters were collected at regular intervals. For the thermal stability (ISOS-D-2), unpackaged perovskite devices were placed on a constant-temperature electric heating stage at 85°C in a N_2 atmosphere, with device parameters recorded periodically. For ISOS-D-3, the encapsulated devices were stored in the dark under controlled external conditions of temperature (85°C) and RH (85%), with periodic measurements of their performance parameters. For device operational stability testing (ISOS-L-2), the packaged perovskite devices were irradiated with light-emitting diode source (PVLIT-6001X-16A) under AM1.5G (1000 W m^{-2}) illumination with calibrated light intensity in N_2 atmosphere. Parameters were collected in real time.

DFT calculations of crystal structure

The DFT calculations were conducted on the basis of the Vienna Ab initio Simulation Package (VASP) (38, 39). The exchange-correlation potential was described by the Perdew-Burke-Ernzerhof (PBE) generalized gradient approach (GGA) (40). The electron-ion interactions were accounted by the projector augmented wave (PAW) (41). The DFT calculations were performed with a cutoff energy of 400 eV, and the $3 \times 2 \times 1$ Gamma centered Monkhorst-Pack grids k-points were selected to sample the Brillouin zone integration. The energy and force convergence criteria of the self-consistent iteration were set to 10^{-5} eV and $0.02 \text{ eV } \text{\AA}^{-1}$, respectively. The DFT-D3 method was used to describe van der Waals interactions (42). Structural optimization and orbitals were performed using the DFT/6-31G(d) with the quantum chemical package Gaussian 09 (43).

Computational methods of ion migration

This study used DFT calculations performed using the VASP (38, 44). The exchange-correlation interactions were described within the GGA framework using the PBE functional (45, 46), while the ion-electron interactions were treated using the PAW method. A plane-wave basis set with a cutoff energy of 520 eV was adopted to ensure computational accuracy. Structural optimizations were conducted with dual convergence criteria: a force convergence threshold of $0.01 \text{ eV } \text{\AA}^{-1}$ and an energy convergence threshold of 10^{-5} eV . The DFT-D3 method with Becke-Johnson damping was incorporated to account for van der Waals interactions, ensuring stable configurations under optimized conditions (38). The Monkhorst-Pack scheme was used to generate k-point meshes, with their densities carefully tested to guarantee a computational error of less than 1 meV per atom.

Structural modeling workflow of ion migration

The initial crystal structures of FAPbI_3 perovskite and its derivatives— $(\text{APD})\text{PbI}_4$, $(4\text{-MPy})\text{PbI}_3$, $(\text{DPE})_{0.5}\text{PbI}_3\cdot\text{DMF}$, and $(\text{APY})_2\text{PbI}_6$ —were first constructed and geometrically optimized to obtain their stable configurations. Subsequently, the optimized derivative structures were used to construct interface models with the FAPbI_3 lattice, resulting in five heterojunction systems: pure FAPbI_3 (110) surface, FAPbI_3 (110)/ $(\text{APD})\text{PbI}_4$ (002) heterojunction, FAPbI_3 (110)/(4-MPy) PbI_3 (011) heterojunction, FAPbI_3 (110)/(DPE) $_{0.5}\text{PbI}_3\cdot\text{DMF}$ (200) heterojunction, and FAPbI_3 (110)/(APY) $_2\text{PbI}_6$ (002) heterojunction. All interface models were fully relaxed until the aforementioned convergence criteria were satisfied.

Migration barrier calculation methodology

To quantify FA^+/MA^+ diffusion kinetics, migration barriers were calculated using the CI-NEB method (47). For each heterojunction, six intermediate images were generated between the fixed initial and final states along the proposed migration pathway. The images were optimized using a conjugate gradient algorithm with spring forces to maintain an approximately uniform image spacing, yielding the minimum energy pathway and the corresponding saddle point. Notably, the calculated barriers represent the total energy change of the entire supercell during coordinated migration. DFT calculations were primarily used for relative comparison across different heterojunction architectures to assess the comparative effectiveness of dimensional confinement and ligand interactions in suppressing cation migration.

Supplementary Materials

The PDF file includes:

Supplementary Text

Figs. S1 to S56

Tables S1 to S14

Legend for movie S1

Legend for data S1

References

Other Supplementary Material for this manuscript includes the following:

Movie S1

Data S1

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